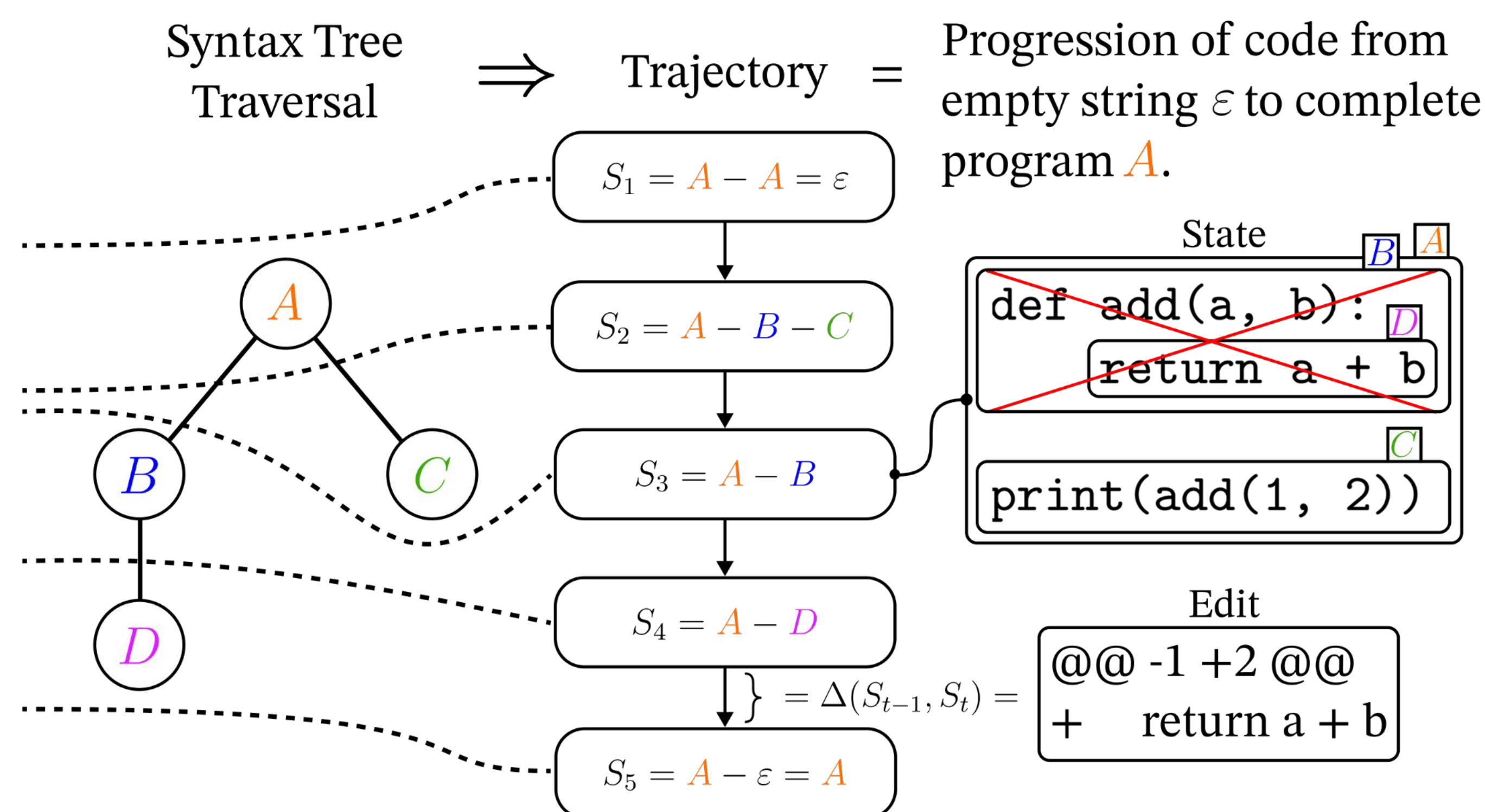


TL;DR: An Approach to Synthesizing Code Trajectories

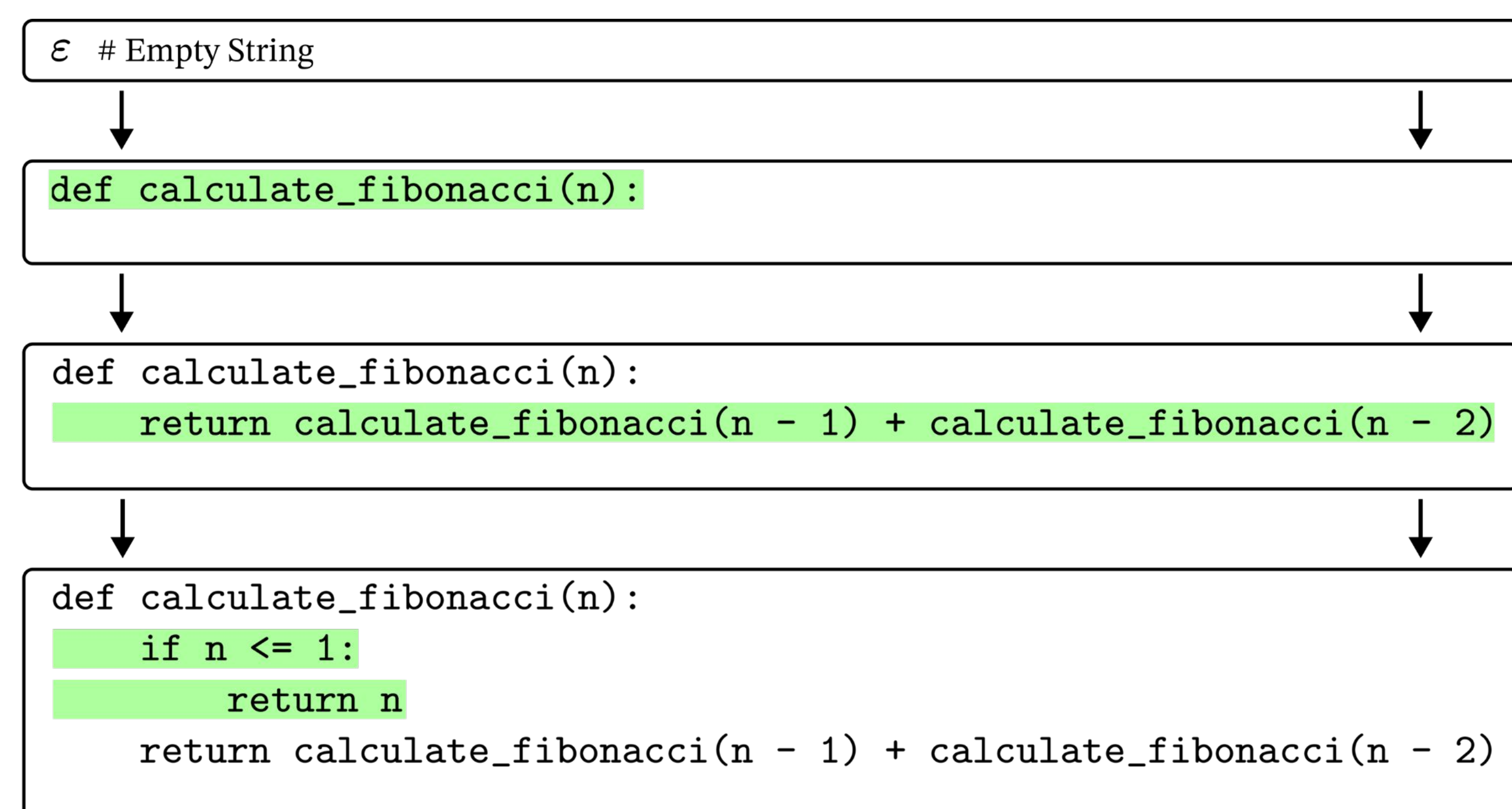
Motivation

- The InstructCoder paper shows Git history trajectories are suboptimal [1].
- The LintSeq paper succeeds while covering only a small part of the data generation space [2].
- No many other effective methods exist for obtaining edit trajectories.
- A parameterized synthesis method can systematically explore optimal edit granularity for pre-training and fine-tuning.
- A well-designed dataset can also ground further research, such as:
 - What is the optimal I/O format for edit modeling?
 - Do multi-turn predictions outperform generating the entire edit path at once?

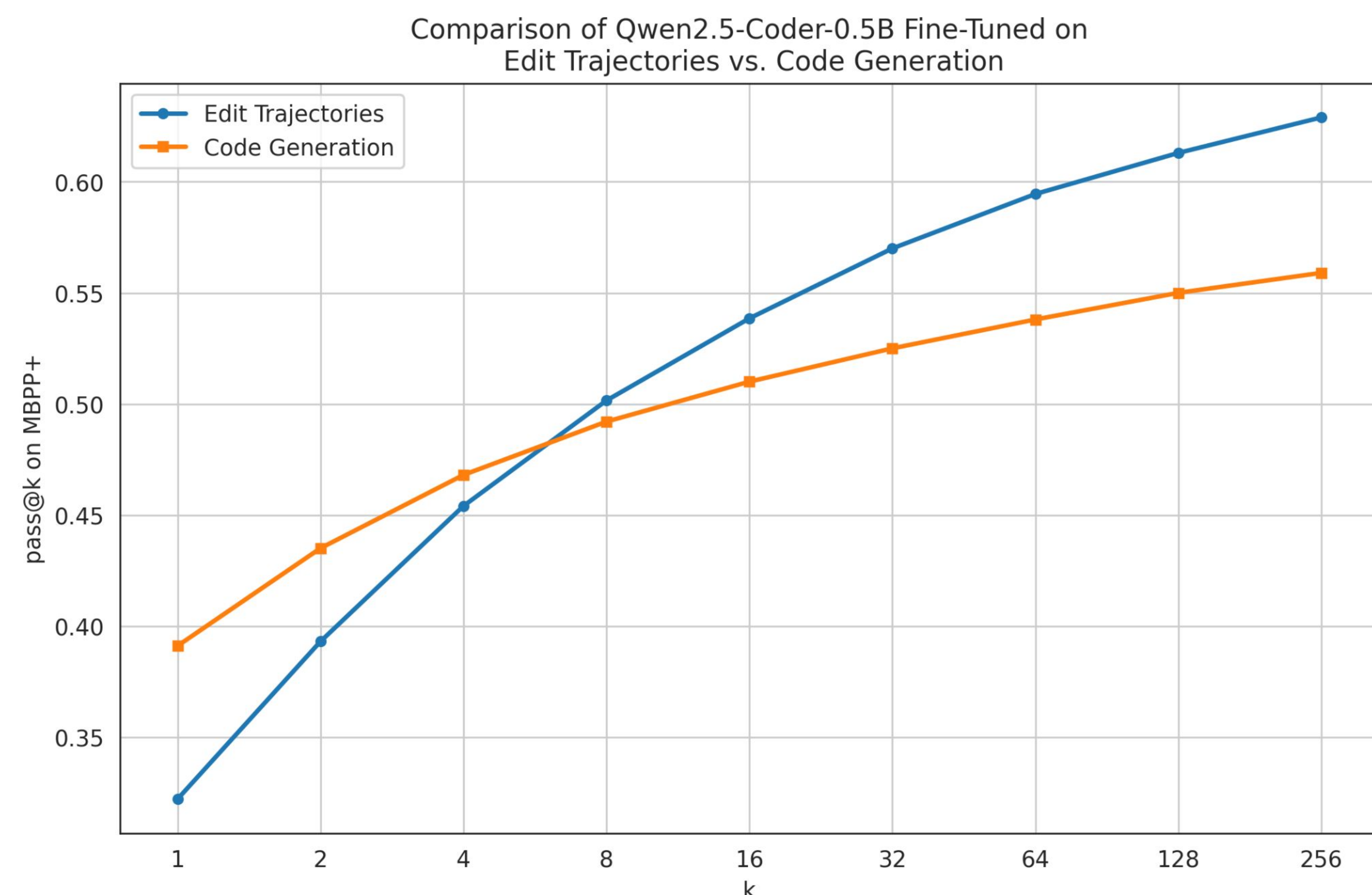
Main Idea



Example of Insertion-Only Trajectory



Evaluation Results from llm-eval



References

- [1] Kaixin Li, Qisheng Hu, Xu Zhao, Hui Chen, Yuxi Xie, Tiedong Liu, Qizhe Xie, & Junxian He. (2024). InstructCoder: Instruction Tuning Large Language Models for Code Editing.
- [2] Ulyana Piterbarg, Lerrel Pinto, & Rob Fergus. (2025). Training Language Models on Synthetic Edit Sequences Improves Code Synthesis.

YT Page



TL;DR: A Benchmark for Exploring Diff Formats

Tasks

$$X - Y = Z$$

$$\text{new_code} - \text{old_code} = \text{diff}$$

Apply

Anti
ApplyDiff
Generation

Diff Formats

```
@@ -1,2 +1,4 @@  
+import math  
+  
def calculate_area(radius):  
-    return 3.14159 * radius * radius  
+    return math.pi * radius * radius  
@@ -4,2 +6,0 @@  
-def old_function():  
-    return "deprecated"
```

```
@@ -1,2 +1,4 @@  
ADD import math  
ADD  
CON def calculate_area(radius):  
DEL     return 3.14159 * radius * radius  
ADD     return math.pi * radius * radius  
@@ -4,2 +6,0 @@  
DEL def old_function():  
DEL     return "deprecated"
```

```
@@ ... @@  
+import math  
+  
def calculate_area(radius):  
-    return 3.14159 * radius * radius  
+    return math.pi * radius * radius  
@@ ... @@  
-def old_function():  
-    return "deprecated"
```

```
<<<<<< SEARCH  
    return 3.14159 * radius * radius  
=====
```

```
<<<<<< SEARCH  
def old_function():  
    return "deprecated"  
=====
```

```
>>>>>> REPLACE
```

Results

Model	Diff Format	Apply	Anti-Apply	Diff Generation		
		EM	EM	EM	f+	f-
GPT 4o	udiff	0.86	0.85	0.43	0.89	0.83
	udiff-h	0.85	0.89	0.03	0.20	0.61
	udiff-l	0.82	0.88	0.03	0.11	0.25
	search-replace	0.57	0.60	0.74	0.93	0.89
GPT 4.1	udiff	0.90	0.88	0.77	0.92	0.91
	udiff-h	0.92	0.93	0.06	0.25	0.61
	udiff-l	0.91	0.88	0.06	0.21	0.39
	search-replace	0.57	0.56	0.95	0.97	0.94
GPT 4.1-mini	udiff	0.89	0.81	0.73	0.87	0.84
	udiff-h	0.86	0.85	0.03	0.28	0.62
	udiff-l	0.86	0.83	0.04	0.10	0.14
	search-replace	0.57	0.55	0.90	0.94	0.94
GPT 4.1-nano	udiff	0.37	0.02	0.51	0.81	0.78
	udiff-h	0.32	0.01	0.12	0.47	0.49
	udiff-l	0.44	0.04	0.09	0.10	0.11
	search-replace	0.33	0.01	0.05	0.07	0.07
Claude 4 Sonnet	udiff	0.95	0.82	0.82	0.95	0.94
	udiff-h	0.95	0.87	0.02	0.13	0.31
	udiff-l	0.95	0.93	0.15	0.34	0.55
	search-replace	0.57	0.48	0.94	0.97	0.96
Gemini 2.5 Flash	udiff	0.90	0.34	0.73	0.94	0.92
	udiff-h	0.91	0.54	0.10	0.22	0.28
	udiff-l	0.79	0.07	0.17	0.32	0.26
	search-replace	0.57	0.53	0.88	0.96	0.95
Qwen2.5-Coder-0.5B	udiff	0.01	0.01	0.00	0.04	0.02
	udiff-h	0.01	0.01	0.03	0.18	0.18
	udiff-l	0.00	0.01	0.23	0.57	0.42
	search-replace	0.00	0.00	0.00	0.04	0.22
Qwen2.5-Coder-1.5B	udiff	0.16	0.04	0.01	0.07	0.09
	udiff-h	0.16	0.05	0.20	0.45	0.42
	udiff-l	0.11	0.03	0.38	0.49	0.49
	search-replace	0.20	0.07	0.19	0.35	0.38
Qwen2.5-Coder-3B	udiff	0.38	0.17	0.01	0.20	0.22
	udiff-h	0.41	0.20	0.02	0.16	0.21
	udiff-l	0.34	0.13	0.36	0.54	0.55
	search-replace	0.28	0.25	0.14	0.30	0.22
Qwen2.5-Coder-7B	udiff	0.58	0.65	0.05	0.39	0.31
	udiff-h	0.57	0.64	0.01	0.17	0.28
	udiff-l	0.57	0.62	0.07	0.17	0.18
	search-replace	0.50	0.50	0.28	0.62	0.58
Qwen2.5-Coder-32B	udiff	0.84	0.87	0.23	0.73	0.63
	udiff-h	0.83	0.86	0.00	0.25	0.67
	udiff-l	0.83	0.82	0.03	0.18	0.39
	search-replace	0.57	0.53	0.68	0.92	0.86

